

Systematic Uncertainty Reduction in Medical AI

A Teach-Back on the Scientific Workflow & Researcher Mindset

Applying disciplined diagnostic reasoning to cross-site AUROC degradation.



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Distinguishing Research Activities

Learning

Learning involves understanding existing knowledge, integrating insights, and developing critical analysis skills to enhance future research applications.

Experimentation

Experimentation entails conducting controlled observations to test hypotheses, yielding data that informs conclusions and drives further inquiry in research.

Engineering

Engineering focuses on creating functioning artifacts or solutions, transforming theoretical concepts into practical applications that solve real-world problems.

Research

Research involves systematic investigation to produce defensible evidence that reduces uncertainty about meaningful questions and contributes to wider knowledge.

The Catalyst: A Cross-Site Degradation

Dataset: Hospital A (Internal Test)

Metric: AUROC

Result: 0.94



Δ -0.21

Dataset: Hospital B (External Test)

Metric: AUROC

Result: 0.73

The model's performance dropped by 0.21. At this stage, we have no metadata, no acquisition protocols, and no demographic data. We only have a mathematical gap.

Restraint Step 1: Separating Observation from Interpretation

What we actually know (Observation)

The model's discrimination and ranking performance on the Hospital B external dataset is 0.21 lower than on the Hospital A internal dataset.

What we assume (Interpretation)

~~The model failed due to distribution shift. It learned the wrong features because the hospitals are different.~~

A drop in AUROC is a measurement.
It is not an explanation.

Deep Dive: Reframing AUROC as a Ranking Mechanism

Concept 1

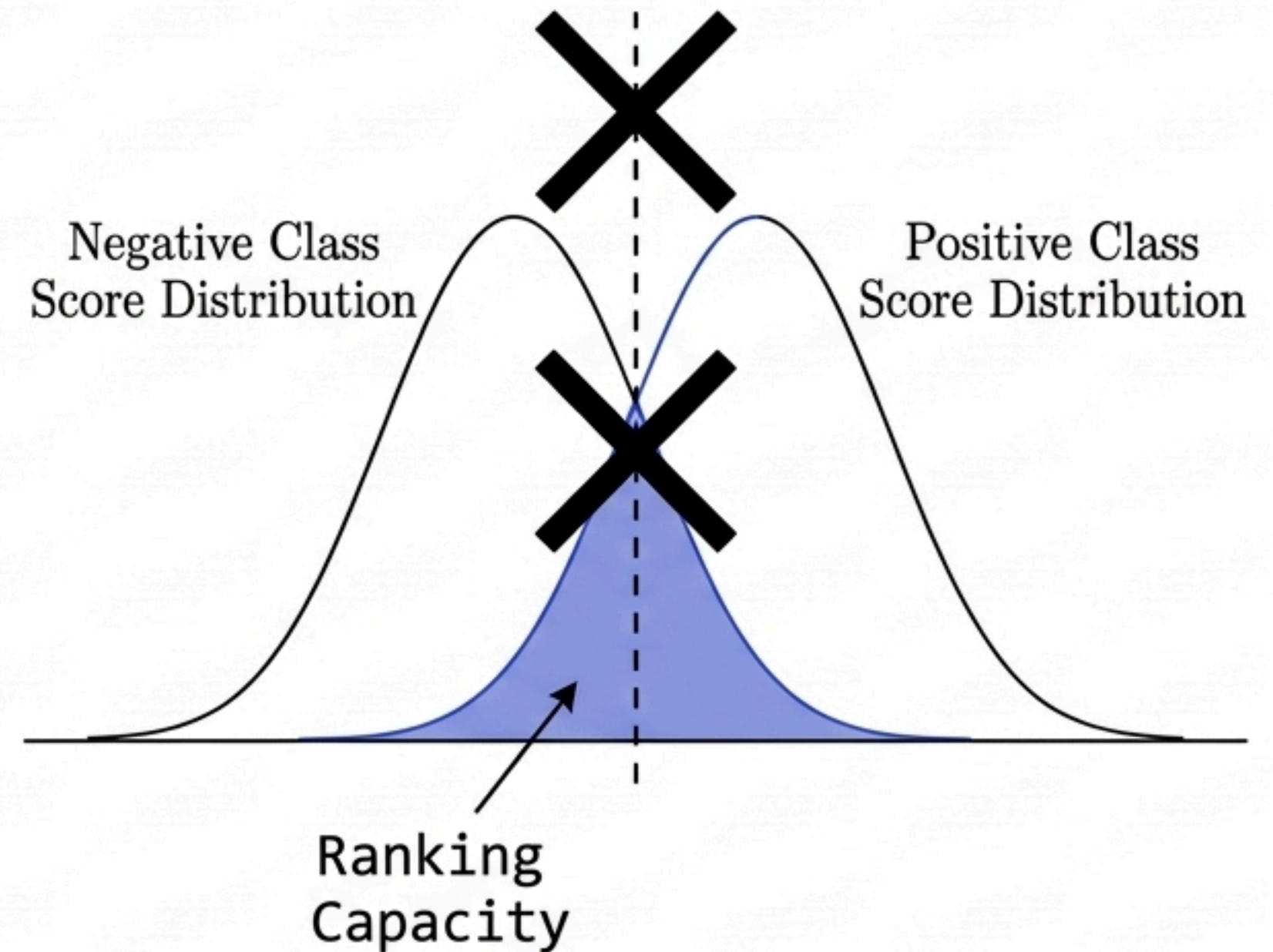
AUROC is NOT accuracy. It evaluates discrimination across all thresholds, measuring how well the model ranks positive examples above negative ones.

Concept 2

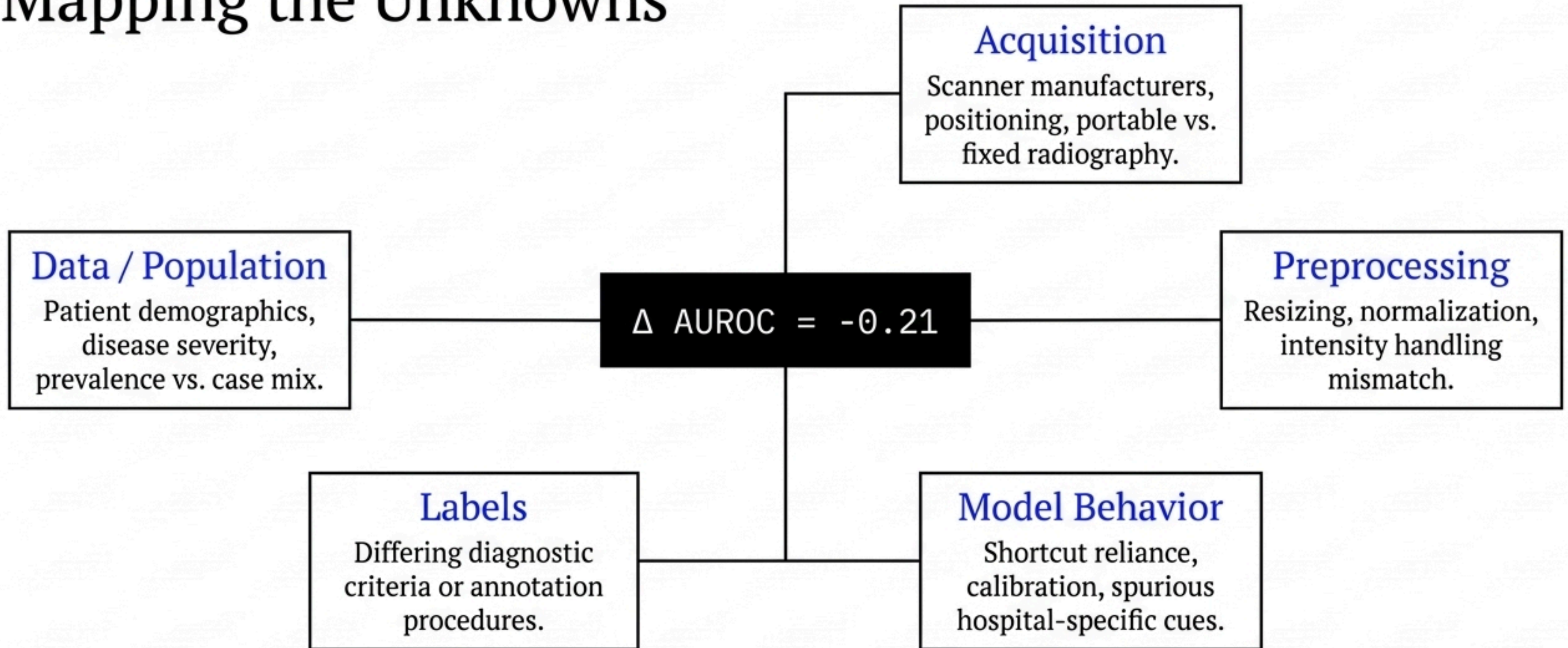
Changing a single decision threshold does not change the AUROC metric.

Research Implication

The Hospital A to B drop cannot be explained by a simple threshold mismatch. The underlying score distributions—the model's fundamental ranking capacity—have shifted.



Embracing Uncertainty: Mapping the Unknowns



Insight: Explicitly listing unknowns prevents jumping to unsupported technical solutions.

Formulating Competing Hypotheses

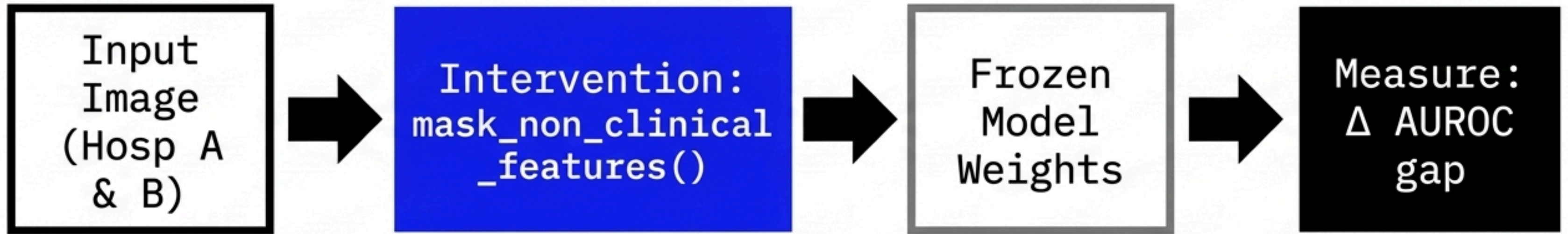
Hypothesis	Mechanism	Cheap Check
H1 (Acquisition Shift)	Differences in scanners/protocols alter image characteristics.	Compare scanner metadata and DICOM headers.
H4 (Preprocessing Mismatch)	Differences in normalization/cropping distort input representation.	Compare dimension and intensity pipelines.
H5 (Shortcut Reliance)	The model relies on non-clinical, hospital-specific features that do not generalize.	Inspect candidate non-clinical features and test them with controlled perturbation/masking.

Focusing the Lens: The Research Question

To what extent does reliance on hospital-specific, non-clinical image features contribute to the reduced external AUROC when evaluated on Hospital B data, compared with its internal performance on Hospital A data?

☒	Why this works:	Identifies a specific mechanism
☒		• Specifies evaluation setting
☒		• Falsifiable by realistic evidence

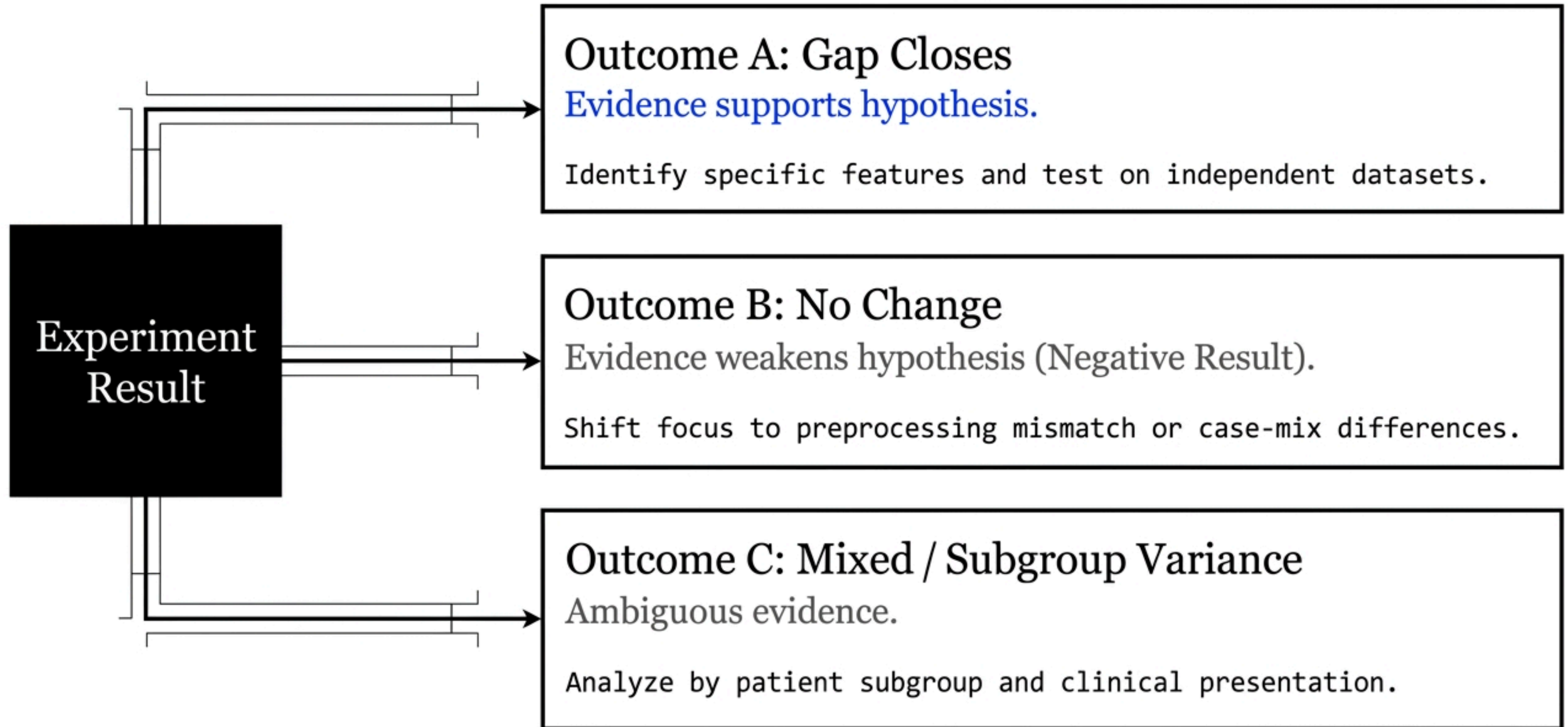
Designing the Smallest Diagnostic Experiment



The Variable: The targeted disruption of suspected non-clinical features.

The Control: Model weights and evaluation procedure remain strictly fixed.

Interpreting Outcomes as Evidence



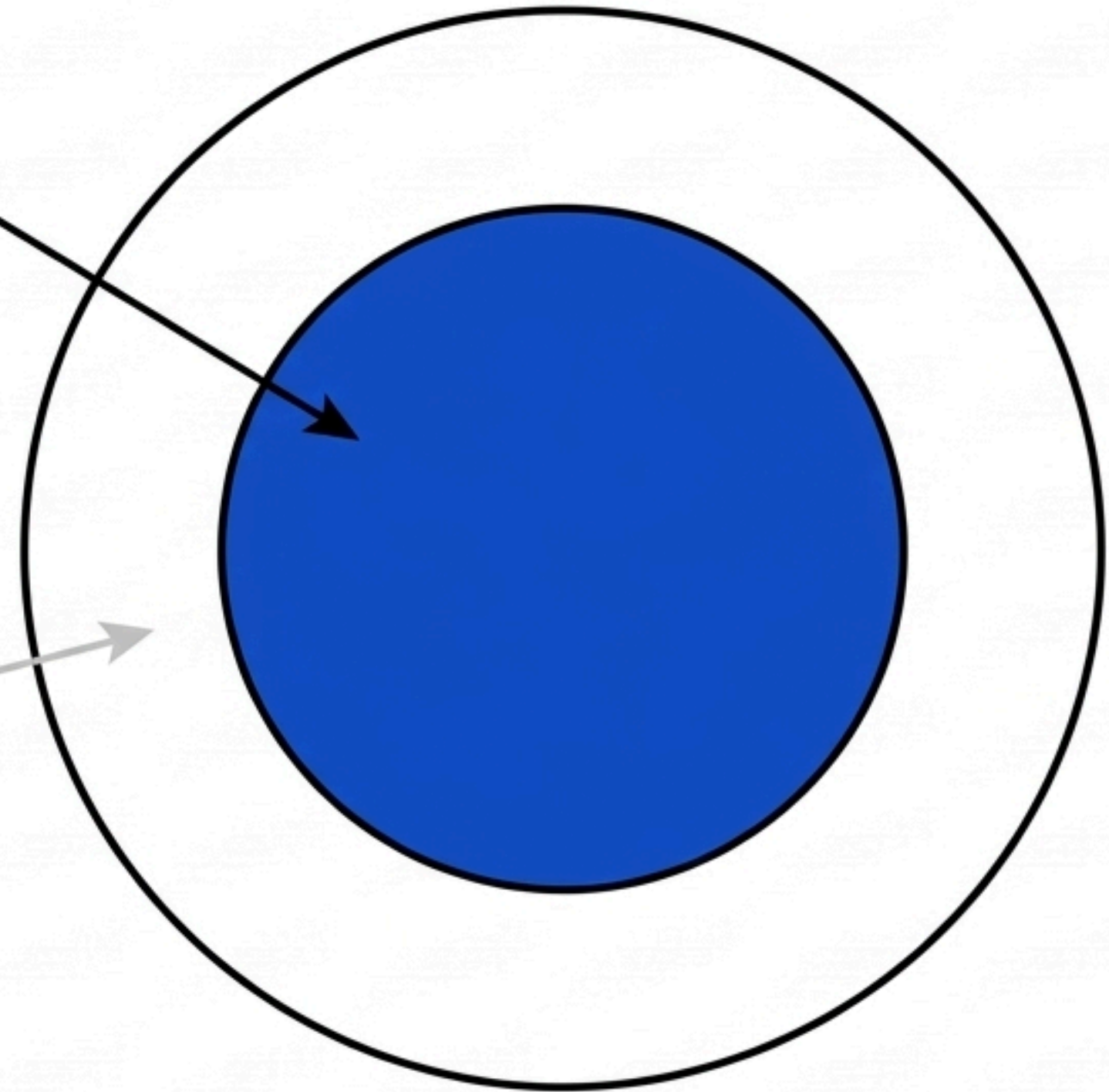
Scientific Restraint: Bounding the Claim

The Bounded Claim

“In one experimental run, disrupting targeted non-clinical features increased Hosp B AUROC from 0.73 to 0.79, suggesting partial shortcut reliance.”
(Proportional to evidence scope)

~~The Overconfident Claim~~

~~“Removing shortcuts solves the distribution shift problem and makes the model clinically robust.”~~
(Exceeds evidence scope)



RULE: CLAIMS MUST REMAIN STRICTLY WITHIN THE SCOPE OF THE POPULATION, INTERVENTION, COMPARISON, AND METRIC TESTED.

Synthesis: The Iterative Research Loop

